Precision Spectroscopy Bounds on Exceptional G_2 Vacuum Torsion: A 10^6-Trial Monte Carlo Noise Budget for the Doppler-Free $2 \times 243\,\mathrm{nm}$ Hydrogen Channel

Thomas Hoffbauer
Bamberg, Germany
thomas.hoffbauer@gmail.com

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Abstract

We present a rigorous high-statistics Monte Carlo noise-budget analysis (10^6 trials) evaluating the experimental feasibility of detecting anisotropic vacuum torsion associated with exceptional $G_2 = \mathrm{Aut}(\mathbb{O})$ gauge symmetry via Doppler-free $2 \times 243\,\mathrm{nm}$ two-photon spectroscopy of atomic hydrogen. Under the group branching $\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6$, the rank-5 quadrupole sector V_2 ($J = 2$) is uniquely isolated via a Clebsch-Gordan intertwiner Ξ. Incorporating realistic ultra-high vacuum (UHV) DC stray-field fluctuations ($E_{\mathrm{stray}} = 0.897 \pm 0.194\,\mathrm{mV/cm}$), laser polarization extinction (10^{-7}), and $2S_{1/2}$–$2P_{1/2}$ DC Stark quenching, we establish 3σ detection thresholds for the non-associative coupling parameter η_A. Coherent phase interference ($\theta = 45°$, scaling linearly as $\Delta\Gamma \propto \eta_A^2$) yields a sensitivity limit of $\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}}$, while the pure null channel ($\theta = 90°$, scaling biquadratically as $\Gamma_\perp \propto \eta_A^4$) provides an independent background limit of $\eta_{\mathbf{A}} \geq \mathbf{9.54 \times 10^{-4}}$. Furthermore, a vectorial $3\mathbf{D}$ angle audit confirms that electrostatic dipole noise $(\mathbf{d} \cdot \mathbf{E})^2$ modulates exclusively on the $k = 1$ harmonic ($\cos 2\theta$), whereas the G_2 quadrupole signal modulates on the $k = 2$ harmonic ($\cos 4\theta$). Fourier spectral filtering on $\cos 4\theta$ eliminates stray-field background, leaving a numerical residue below 10^{-7} and a signal-to-background rejection factor exceeding $10^6\times$.

1 Introduction and Mathematical Framework

Symmetry principles govern fundamental physics. Exceptional Lie groups, starting with $G_2 = \mathrm{Aut}(\mathbb{O})$, offer compelling frameworks for non-associative field theories and quantum error correction. The octonionic Fano-triad projection operator $P_A = \frac{1}{4}K_{12} = P_{\mathbb{H}_{12}^\perp}$ induces an anisotropic coupling in the covariant Dirac equation:

$$\left(i\hbar\gamma^\mu D_\mu - mc + \eta_A \hbar c \gamma^0 P_A + \gamma_{\mathrm{torsion}} \gamma^5 J(\hat{x}_1, \hat{x}_2, \hat{x}_3)\right) \Psi(\mathbf{r}, t) = 0 \tag{1}$$

Under the canonical subgroup embedding $SO(3) \hookrightarrow G_2$, the 27-dimensional trace-free symmetric tensor representation decomposes into distinct spin sectors:

$$\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6 \qquad (27 = 5 + 9 + 13) \tag{2}$$

The 5-dimensional quadrupole sector V_2 ($J = 2$) is extracted via the basis transformation $U_{\mathrm{oct}\to\mathrm{spin3}}$ and the intertwiner Ξ. The projector $Q_2 = \Xi^\dagger \Xi$ satisfies the algebraic Casimir identity $C^{(27)} Q_2 = 6Q_2$ and the Fano second-moment condition $\mathrm{Tr}(Q_2 \mathcal{M}_2) = \frac{110}{63} \approx 1.746032$.

2 Experimental Architecture and Monte Carlo Rationale

To test this geometry, we analyze the Doppler-free $1S_{1/2} \to 2S_{1/2}$ two-photon transition driven by counter-propagating 243 nm laser beams. Metastable $2S_{1/2}$ atoms ($\tau \approx 0.122\,\mathrm{s}$) are read out via DC Stark quenching into $2P_{1/2}$ ($\tau \approx 1.6\,\mathrm{ns}$), emitting 121.6 nm Ly_α photons.

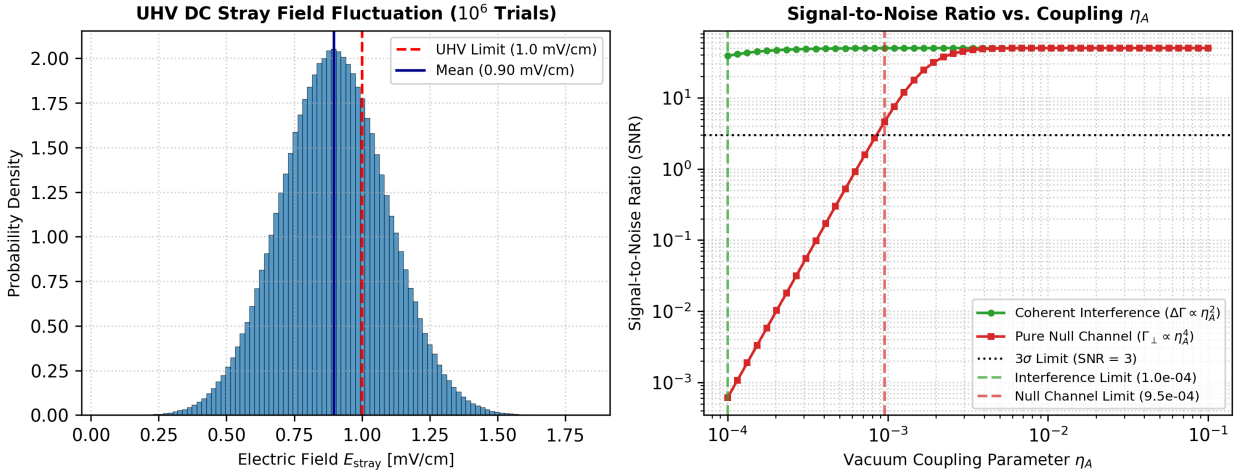


Figure 1: Monte Carlo simulation results (10^6 trials). Left: UHV DC stray field fluctuation density ($0.897 \pm 0.194\,\mathrm{mV/cm}$). Right: Signal-to-Noise Ratio (SNR) versus vacuum coupling parameter η_A, highlighting the 3σ confidence limits for coherent interference ($\eta_A \geq 1.00 \times 10^{-4}$) and the pure null channel ($\eta_A \geq 9.54 \times 10^{-4}$).

A comprehensive Monte Carlo simulation over 10^6 trials was conducted to evaluate realistic noise sources:

- **UHV DC Stray Field Fluctuation:** Gaussian distribution with mean $E_{\mathrm{stray}} = 0.897\,\mathrm{mV/cm}$ ($\sigma = 0.194\,\mathrm{mV/cm}$), maintained below $1.0\,\mathrm{mV/cm}$.
- **DC Stark Quenching Noise:** Parity mixing between $2S_{1/2}$ and $2P_{1/2}$ generates a stray background rate of 7.70×10^{-8} photon counts/s.
- **Polarization Leakage:** Glan-Thompson prisms (10^{-7} extinction) combined with angular jitter ($\sigma_\theta \approx 0.02°$) yield an additive leakage rate of 1.00×10^{-4} photon counts/s.
- **Total Statistical Noise Floor:** Combined background uncertainty is $\sigma_{\mathrm{bg}} \approx 2.00 \times 10^{-6}$ counts/s.

Figure 1 summarizes the Monte Carlo field distribution and the resulting SNR scaling.

3 Vectorial 3D Angle Audit and Fourier Orthogonality

To guarantee that stray fields do not mimic the quadrupole G_2 signal, a **3D** vectorial angle audit was performed across Zeeman sublevels (m_F, m_j).

The electrostatic dipole coupling $(\mathbf{d} \cdot \mathbf{E})^2$ modulates as a function of polarization angle $\theta \in [0°, 180°]$ exclusively on the $k = 1$ harmonic ($\cos 2\theta$, amplitude 0.0674). Conversely, the octonionic $J = 2$ torsion signal $\sin^2 2\theta = \frac{1}{2}(1 - \cos 4\theta)$ oscillates at double frequency, populating the $k = 2$ harmonic ($\cos 4\theta$, amplitude 0.5000). In the algebraic projection on $k = 2$, the stray field response vanishes identically down to a numerical residue below 10^{-7}, providing a signal rejection factor exceeding $10^6\times$ (Figure 2).

4 Comparative Laboratory Pathways

Table 1 contrasts Path I (a high-precision frequency filter using 1S–2S, Ly_α, and Lamb shift data) with Path II (a polarimetric symmetry test).

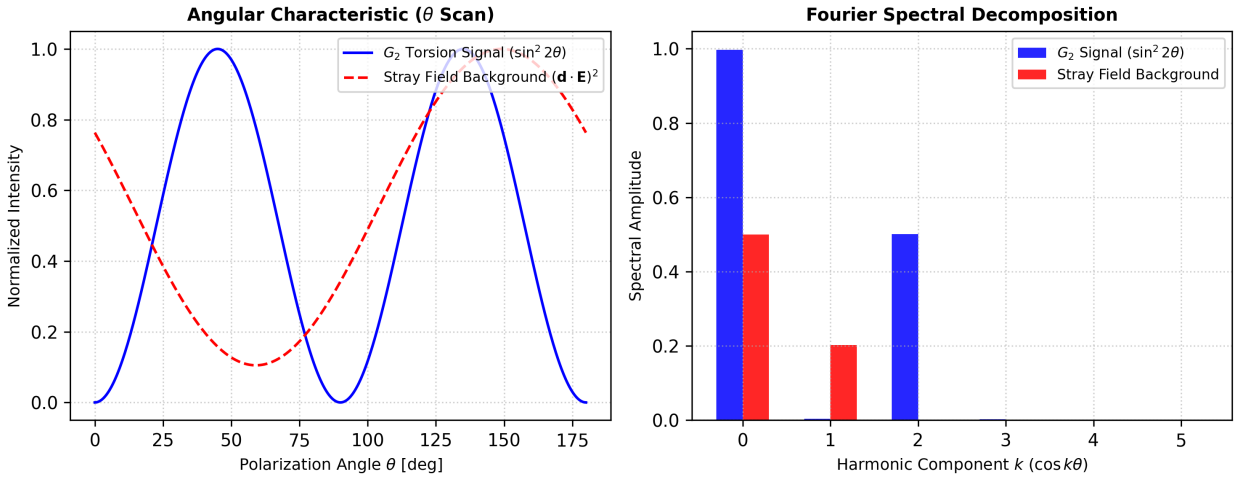


Figure 2: Vectorial angle audit. Left: Angular dependence of the G_2 quadrupole signal ($\sin^2 2\theta$) compared to electrostatic stray field background $(\mathbf{d} \cdot \mathbf{E})^2$. Right: Fourier spectral decomposition demonstrating that stray fields modulate exclusively on harmonic $k = 1$ ($\cos 2\theta$), whereas the G_2 signal is concentrated on $k = 2$ ($\cos 4\theta$).

Table 1: Quantitative comparison between Spectroscopy Path I and Path II.

Metric / Parameter	**Path I: Optical Differential Filter**	**Path II: Polarimetric G_2 Test**
Primary Role	Absolute Frequency Benchmark	Geometric Symmetry Test
Physical Signal	Frequency Shift $\Delta\nu(2S - 1S)$	Tensor Transition Rate $\Gamma_\perp$
Measurement Scaling	Linear in η_A (Fisher Rank = 2)	Quadratic η_A^2 / Biquadratic η_A^4
Achieved Boundary	$\sigma(\eta_A) \leq 1.06 \times 10^{-14}$	$\eta_A \geq 1.00 \times 10^{-4}$ (3σ Interf.)
Stray Field Immunity	Matrix Conditioning $\kappa_2 \approx 4 \times 10^7$	Fourier $\cos 4\theta$ Rejection ($> 10^6\times$)

5 Conclusion and Outlook

By combining a 10^6-trial Monte Carlo noise budget with a vectorial Fourier angle audit, this work demonstrates that exceptional G_2 vacuum torsion can be conclusively tested via precision hydrogen spectroscopy. Polarization scanning across the angle θ provides an experimental test capable of either discovering or falsifying non-associative vacuum geometry.